KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

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SCHOOL OF ELECTRONICS

PROJECT

VEHICLE MANAGEMENT SYSTEM USING JAVA

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📌 PROBLEM STATEMENT

In real life, managing vehicle information such as cars and bikes is difficult if done manually.
There is a need for a simple software system that can:

- Manage different types of vehicles
- Store vehicle details
- Allow comparison between vehicles
- Be easy to use and menu-driven

Traditional systems do not effectively demonstrate object-oriented programming concepts such as inheritance and polymorphism.

🎯 OBJECTIVES OF THE PROJECT

The main objectives of this project are:

1. To develop a menu-driven Vehicle Management System using Java
2. To implement **inheritance** using a parent Vehicle class
3. To implement **polymorphism** by handling Car and Bike objects together
4. To allow users to:
    - Configure cars and bikes
    - Save vehicle history
    - Compare two vehicles of the same type
5. To understand real-world application of Java OOP concepts

🧠 CONCEPTS USED IN JAVA

This project uses the following Java concepts:

- **Inheritance** – Car and Bike inherit from Vehicle
- **Abstraction** – Vehicle is an abstract class
- **Polymorphism** – Vehicle reference stores Car and Bike objects
- **Encapsulation** – Data members are private/protected
- **ArrayList** – To store multiple vehicles dynamically
- **Scanner** – For user input
- **Menu-driven programming**

🏗️ SYSTEM DESIGN (OVERVIEW)

- `Vehicle` → Abstract parent class
- `Car` → Child class
- `Bike` → Child class
- `VehicleManagement` → Main class (menu & control)

💻 JAVA SOURCE CODE

```
import java.util.*;

// ================= PARENT CLASS =================
abstract class Vehicle {
    protected String type;
    protected String company;
    protected String model;
    protected String fuel;

    abstract void configure(Scanner sc);
    abstract void display();
}

// ================= CAR CLASS =================
class Car extends Vehicle {
    private String range;
    private String seats;
    private String sunroof;

    public Car() {
        type = "Car";
    }

    public void configure(Scanner sc) {
        System.out.println("\nChoose Car Company:");
        System.out.println("1. Maruti\n2. Hyundai\n3. Tata");
        int c = sc.nextInt();
        company = (c == 1) ? "Maruti" : (c == 2) ? "Hyundai" : "Tata";

        System.out.println("Choose Model:");
        System.out.println("1. Base\n2. Mid\n3. Top");
        int m = sc.nextInt();
        model = (m == 1) ? "Base" : (m == 2) ? "Mid" : "Top";

        System.out.println("Choose Price Range:");
        System.out.println("1. 5–10 Lakh\n2. 10–15 Lakh\n3. 15–20 Lakh");
        int r = sc.nextInt();
        range = (r == 1) ? "5–10 Lakh" : (r == 2) ? "10–15 Lakh" : "15–20 Lakh";

        System.out.println("Choose Seats:");
        System.out.println("1. 4\n2. 5");
        int s = sc.nextInt();
        seats = (s == 1) ? "4" : "5";
```

```java
        System.out.println("Choose Fuel:");
        System.out.println("1. Petrol\n2. Diesel\n3. Electric");
        int f = sc.nextInt();
        fuel = (f == 1) ? "Petrol" : (f == 2) ? "Diesel" : "Electric";

        System.out.println("Sunroof:");
        System.out.println("1. Yes\n2. No");
        int sr = sc.nextInt();
        sunroof = (sr == 1) ? "Yes" : "No";

        System.out.println("✔ Car saved successfully!\n");
    }

    public void display() {
        System.out.println("Type     : Car");
        System.out.println("Company  : " + company);
        System.out.println("Model    : " + model);
        System.out.println("Range    : " + range);
        System.out.println("Seats    : " + seats);
        System.out.println("Fuel     : " + fuel);
        System.out.println("Sunroof  : " + sunroof);
    }
}

// ================= BIKE CLASS =================
class Bike extends Vehicle {
    private String engine;
    private String abs;

    public Bike() {
        type = "Bike";
    }

    public void configure(Scanner sc) {
        System.out.println("\nChoose Bike Company:");
        System.out.println("1. Hero\n2. Honda\n3. Yamaha");
        int c = sc.nextInt();
        company = (c == 1) ? "Hero" : (c == 2) ? "Honda" : "Yamaha";

        System.out.println("Choose Model:");
        System.out.println("1. Standard\n2. Sports\n3. Premium");
        int m = sc.nextInt();
        model = (m == 1) ? "Standard" : (m == 2) ? "Sports" : "Premium";

        System.out.println("Choose Engine CC:");
        System.out.println("1. 100–125cc\n2. 150–160cc\n3. 200+cc");
```

```java
        int e = sc.nextInt();
        engine = (e == 1) ? "100–125cc" : (e == 2) ? "150–160cc" : "200+cc";

        System.out.println("Choose Fuel:");
        System.out.println("1. Petrol\n2. Electric");
        int f = sc.nextInt();
        fuel = (f == 1) ? "Petrol" : "Electric";

        System.out.println("ABS:");
        System.out.println("1. Yes\n2. No");
        int a = sc.nextInt();
        abs = (a == 1) ? "Yes" : "No";

        System.out.println("✔ Bike saved successfully!\n");
    }

    public void display() {
        System.out.println("Type     : Bike");
        System.out.println("Company  : " + company);
        System.out.println("Model    : " + model);
        System.out.println("Engine   : " + engine);
        System.out.println("Fuel     : " + fuel);
        System.out.println("ABS      : " + abs);
    }
}


// ================= MAIN CLASS =================
public class VehicleManagement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Vehicle> vehicles = new ArrayList<>();

        while (true) {
            System.out.println("\n================================");
            System.out.println(" VEHICLE MANAGEMENT SYSTEM");
            System.out.println("================================");
            System.out.println("1. Configure Car");
            System.out.println("2. Configure Bike");
            System.out.println("3. Show Saved Vehicles");
            System.out.println("4. Compare Vehicles");
            System.out.println("5. Exit");
            System.out.print("Enter choice: ");

            int choice = sc.nextInt();

            switch (choice) {
```

```java
        case 1:
            Vehicle car = new Car();
            car.configure(sc);
            vehicles.add(car);
            break;

        case 2:
            Vehicle bike = new Bike();
            bike.configure(sc);
            vehicles.add(bike);
            break;

        case 3:
            System.out.println("\n====== SAVED VEHICLES ======");
            for (int i = 0; i < vehicles.size(); i++) {
                System.out.println("\nIndex: " + i);
                vehicles.get(i).display();
            }
            break;

        case 4:
            System.out.print("Enter first vehicle index: ");
            int i1 = sc.nextInt();
            System.out.print("Enter second vehicle index: ");
            int i2 = sc.nextInt();

            if (i1 >= vehicles.size() || i2 >= vehicles.size()) {
                System.out.println("Invalid index!");
            } else if (!vehicles.get(i1).type.equals(vehicles.get(i2).type)) {
                System.out.println("Cannot compare different vehicle types!");
            } else {
                System.out.println("\n===== COMPARISON =====");
                vehicles.get(i1).display();
                System.out.println("---------------------");
                vehicles.get(i2).display();
            }
            break;

        case 5:
            System.out.println("Thank you for using Vehicle Management System.");
            sc.close();
            return;

        default:
            System.out.println("Invalid choice!");
    }
}
```

```
    }
}
```

📌 SAMPLE OUTPUT

Project: Vehicle Management System using Java

▶️ PROGRAM START

```
C:\Users\KIIT\Desktop\ppj>javac  VehicleManagement.java

C:\Users\KIIT\Desktop\ppj>java  VehicleManagement

==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 1
```

🚗 CONFIGURE FIRST CAR

```
Enter choice: 1

Choose Car Company:
1. Maruti
2. Hyundai
3. Tata
3
Choose Model:
1. Base
2. Mid
3. Top
3
Choose Price Range:
1. 5?10 Lakh
2. 10?15 Lakh
3. 15?20 Lakh
2
Choose Seats:
1. 4
2. 5
2
Choose Fuel:
1. Petrol
2. Diesel
3. Electric
3
Sunroof:
1. Yes
2. No
1
? Car saved successfully!
```

🚗 CONFIGURE SECOND CAR

```
Choose Car Company:
1. Maruti
2. Hyundai
3. Tata
2
Choose Model:
1. Base
2. Mid
3. Top
2
Choose Price Range:
1. 5?10 Lakh
2. 10?15 Lakh
3. 15?20 Lakh
3
Choose Seats:
1. 4
2. 5
1
Choose Fuel:
1. Petrol
2. Diesel
3. Electric
2
Sunroof:
1. Yes
2. No
1
? Car saved successfully!
```

⚖️ COMPARE TWO CARS

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 4
Enter first vehicle index: 0
Enter second vehicle index: 1

===== COMPARISON =====
Type      : Car
Company   : Tata
Model     : Top
Range     : 10?15 Lakh
Seats     : 5
Fuel      : Electric
Sunroof   : Yes
---------------------
Type      : Car
Company   : Hyundai
Model     : Mid
Range     : 15?20 Lakh
Seats     : 4
Fuel      : Diesel
Sunroof   : Yes
```

🏍️ CONFIGURE FIRST BIKE

```
==================================
 VEHICLE MANAGEMENT SYSTEM
==================================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 2

Choose Bike Company:
1. Hero
2. Honda
3. Yamaha
2
Choose Model:
1. Standard
2. Sports
3. Premium
3
Choose Engine CC:
1. 100?125cc
2. 150?160cc
3. 200+cc
2
Choose Fuel:
1. Petrol
2. Electric
2
ABS:
1. Yes
2. No
1
? Bike saved successfully!
```

🏍️ CONFIGURE SECOND BIKE

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 2

Choose Bike Company:
1. Hero
2. Honda
3. Yamaha
1
Choose Model:
1. Standard
2. Sports
3. Premium
2
Choose Engine CC:
1. 100?125cc
2. 150?160cc
3. 200+cc
2
Choose Fuel:
1. Petrol
2. Electric
1
ABS:
1. Yes
2. No
2
? Bike saved successfully!
```

⚖️ COMPARE TWO BIKES

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 4
Enter first vehicle index: 2
Enter second vehicle index: 3

===== COMPARISON =====
Type      : Bike
Company   : Honda
Model     : Premium
Engine    : 150?160cc
Fuel      : Electric
ABS       : Yes
---------------------
Type      : Bike
Company   : Yamaha
Model     : Sports
Engine    : 200+cc
Fuel      : Petrol
ABS       : No
```

📋 SHOW ALL SAVED VEHICLES (HISTORY)

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 3

====== SAVED VEHICLES ======

Index: 0
Type     : Car
Company  : Tata
Model    : Top
Range    : 10?15 Lakh
Seats    : 5
Fuel     : Electric
Sunroof  : Yes

Index: 1
Type     : Car
Company  : Hyundai
Model    : Mid
Range    : 15?20 Lakh
Seats    : 4
Fuel     : Diesel
Sunroof  : Yes
```

```
Index: 2
Type      : Bike
Company   : Honda
Model     : Premium
Engine    : 150?160cc
Fuel      : Electric
ABS       : Yes

Index: 3
Type      : Bike
Company   : Hero
Model     : Sports
Engine    : 150?160cc
Fuel      : Petrol
ABS       : No
```

EXIT PROGRAM

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 5
Thank you for using Vehicle Management System.
```

🧾 CONCLUSION

This project successfully demonstrates the use of Java Object-Oriented Programming concepts in a real-world application.
By using inheritance, abstraction, and polymorphism, the Vehicle Management System efficiently manages different types of vehicles.

The project is user-friendly, scalable, and suitable for academic purposes. It provides a strong foundation for understanding how real applications are built using Java.